General Mathematics — Exercise Sheet 27

Improper integrals and the Gamma function

Variant: Statements

Exercise 1 (Convergence with a real parameter). For each $p \in \mathbb{R}$, determine whether each of the following improper integrals converges. Evaluate the first when it converges; the second requires a convergence test only.

$$\text{(a)} \quad \int_e^{\infty} \frac{\mathrm{d}x}{x(\ln x)^p}, \qquad \text{(b)} \quad \int_0^1 \frac{\sin x}{x^p}\,\mathrm{d}x.$$

Use the power tests of Lecture 27 and $\sin x/x \to 1$ from Lecture 6. Apply substitutions only on finite intervals before taking endpoint limits.

Exercise 2 (Conditional convergence and a tail bound). Prove the following claims about the improper integral of $\sin x/x$ over $[1, \infty)$.

(a) The integral converges. Use integration by parts on $[1, R]$ and absolute comparison with x^{-2}.

(b) For every $R \geq 1$, its convergent tail has absolute value at most $2/R$.

(c) Its absolute-value integral diverges. Use the specific bound $|\sin x| \geq \sin^2 x = (1 - \cos(2x))/2$ and prove that $\int_1^{\infty} \cos(2x)/x\,\mathrm{d}x$ converges.

The derivatives of sine and cosine are from Lecture 8; finite definite parts is from Lecture 26. No value of the sine integral at infinity is assumed.

Exercise 3 (A limit concentrating at an endpoint). Let $f : [0, 1] \to \mathbb{R}$ be continuous. For every integer $n \geq 0$, define

$$A_n = \int_0^1 x^n f(x)\,\mathrm{d}x.$$

Prove $A_n \to 0$ and $(n + 1)A_n \to f(1)$ as $n \to \infty$. Use the extreme value bound from Lecture 7, finite integral order and additivity from Lecture 25, the fundamental theorem from Lecture 26, and the geometric sequence limit from Lecture 3.

Exercise 4 (Wallis integrals and a product limit). For each integer $n \geq 0$, define the *Wallis integral* by

$$W_n = \int_0^{\pi/2} \sin^n x\,\mathrm{d}x,$$

where $\sin^0 x$ means the constant function 1, including at zero.

(a) Compute W_0 and W_1. Use the product rule and the trigonometric derivatives of Lecture 8 to prove $W_n = (n - 1)W_{n-2}/n$ for $n \geq 2$.

(b) Prove the finite-product formulas, for every integer $k \geq 0$,

$$W_{2k} = \frac{\pi}{2} \prod_{j=1}^{k} \frac{2j - 1}{2j}, \qquad W_{2k+1} = \prod_{j=1}^{k} \frac{2j}{2j + 1}.$$

An empty product is 1.

(c) Prove the Wallis product limit

$$\lim_{k\to\infty} \prod_{j=1}^{k} \frac{4j^2}{4j^2-1} = \frac{\pi}{2}.$$

Use $0 \le \sin x \le 1$ on $[0, \pi/2]$, finite integral order, and the sequence squeeze theorem of Lecture 4.

Exercise 5 (A Gamma inequality without the Gaussian value). For $s, t > 0$, prove

$$\Gamma\left(\frac{s+t}{2}\right)^2 \le \Gamma(s)\Gamma(t),$$

and show that equality holds exactly when $s = t$. Deduce $0 < \Gamma(1/2) < 2$ and hence $0 < \int_0^\infty e^{-x^2}\,dx < 1$ without using its value from Lecture 38. Use the Gamma definition, convergence, recurrence and Gaussian relation from Lecture 27; finite integral linearity and order from Lecture 25; and continuity from Lecture 7. Prove the inequality first on finite positive cutoffs by integrating a square, then pass to the separate endpoint limits.